Sparse Biomarker Discovery from Low-Montage Resting-State EEG for tDCS Treatment Response in Adults with Treatment-Resistant Depression

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ABSTRACT

We evaluated whether a sparse, low-montage resting-state EEG pipeline could identify clinically interpretable biomarkers of response to transcranial direct current stimulation (tDCS) in adults with treatment-resistant depression from a fully remote home-based trial, under strict patient-level validation. The study used 4-channel eyes-closed EEG and a controlled sweep of 150 successful runs across two model families: an advanced hybrid 1D CNN-LSTM and a linear support vector machine (SVM). The sweep varied window size (2–10 s) and the per-fold feature-selection budget (`inner-k` 1–70), while enabling both leave-one-participant-out group equalization and SMOTE during training.

The strongest configuration was the hybrid model with 6 s windows and `inner-k=1`. That run reached patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, and MCC 0.571 over 21 held-out participants. The best SVM configuration (8 s, `inner-k=30`) reached ROC-AUC 0.510 and MCC 0.277. The hybrid winner was also highly sparse, selecting 1 feature per fold and only 7 unique features overall, with a permutation-tested ROC-AUC advantage over chance ($p = 0.008$). Hard-set overlap across folds was modest (mean pairwise Jaccard 0.190; Kuncheva index 0.187). The recurrent features showed perfect mean effect-direction consistency.

These results support a sparse short-timescale EEG framework that recovers a small, interpretable biomarker set while preserving useful patient-level discrimination in a relatively small cohort. The most recurrent features involved temporal spectral entropy and frontal-temporal asymmetry-derived measures, and their class-conditional selection patterns motivate follow-up validation in larger externally evaluated datasets.

1. Introduction

Treatment-resistant depression (TRD) remains difficult to manage, and current clinical guidance continues to place neuromodulation among the options considered after inadequate response to standard treatments [1]. Meta-analytic evidence has reinforced the broader clinical relevance of non-invasive brain stimulation in depressive illness [2], and a tDCS-specific meta-analysis reported antidepressant effects across depressive episodes [3]. In major depression, early randomized evidence for tDCS was reported by Boggio et al. [4], and an individual-patient meta-analysis later supported its acute antidepressant efficacy [5]. Practical design and implementation guidance has also helped standardize how tDCS studies are conducted [6], while broader translational neuromodulation work has framed such interventions within a wider neuroplasticity-based clinical agenda [7]. More recently, home-based delivery has moved from concept to clinical trial: a fully remote phase 2 randomized sham-controlled study demonstrated home-based tDCS in major depressive disorder [8], and qualitative follow-up work found the approach acceptable to participants [9]. These advances make pretreatment stratification a practical clinical problem. A useful biomarker pipeline has to identify which patients are more or less likely to respond before stimulation begins.

EEG is well suited to that problem because wearable hardware has made ambulatory recording increasingly feasible [10], broader clinical overviews continue to support diagnostic, prognostic, and therapeutic uses of EEG-based markers [11], and recent technical guidance has emphasized both the promise and the implementation challenges of EEG biomarkers in major depressive disorder [12]. Reduced-channel systems improve portability and ease of deployment [13], whereas denser acquisitions provide richer spatial detail at the cost of greater setup burden [14]. Standard recording practice therefore remains important even in low-montage settings [15].

The literature on EEG biomarkers in major depressive disorder is encouraging and methodologically uneven. Reviews and biomarker overviews have repeatedly highlighted spectral, asymmetry, vigilance, entropy, and complexity signatures as potentially informative marker families [16–18]. Related discussions of neuroimaging and EEG biomarkers have emphasized the clinical opportunity, but also the need for more robust treatment-response models and better translational discipline [19–21]. More recent meta-analytic work on EEG-based treatment prediction reaches a similar conclusion: useful signal is present, but translation depends on stronger methodology and standardization [12, 22, 23]. Machine learning studies have extended this literature into antidepressant, placebo,

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and neuromodulation response prediction, while also exposing recurrent concerns about external generalization, feature stability, leakage-safe selection, and interpretability [24–26]. Related nonlinear and entropy-oriented studies point to the same caution about dimensionality and signal instability [27–29]. Entropy-focused EEG work has reinforced the relevance of complexity measures [30–32]. Leakage-specific and data-centric critiques make the underlying design problem explicit [33, 34].

That tension is especially sharp in relatively small neuromodulation studies. Rich models can improve representation learning, but relatively small cohorts also increase the risk of unstable feature selection, inflated performance estimates, and overly confident biomarker claims. Nested evaluation and within-fold processing are therefore central design issues rather than implementation details in this literature [21, 33]. Clinically deployable EEG biomarkers also need interpretability and low-density feasibility, and recent data-centric frameworks have started to move in that direction [34]. Emerging response-prediction studies in depression and related neuromodulation settings show substantial promise [35–37]. Related reviews and pilot studies also make the methodological fragility clear [38–40]. More recent response-oriented studies continue that pattern [41–43]. A recent systematic review of EEG connectivity changes in depression treatment reaches a similar conclusion [44]. Within the same home-based 4-channel MDD program studied here, recent PSD-based and PLV-based analyses have already shown that the baseline EEG carries treatment-response information across multiple signal representations [45, 46]. That background makes this cohort a useful setting for testing whether a sparse interpretable feature-selection workflow can isolate a small recurrent biomarker set under strict participant-level validation.

This study focuses on a narrow objective: sparse biomarker discovery under strict participant-level evaluation. We compare two classifiers over the same feature-selection sweep to test whether a low-montage resting-state EEG pipeline can recover a small recurring set of candidate markers associated with tDCS response in adults with TRD. The emphasis is on parsimony, candidate-biomarker interpretability, and leakage-resistant validation. Our working hypothesis was that useful patient-level discrimination could be achieved with a small per-fold feature budget and that the most recurrent features would show stronger effect-direction consistency than hard-set overlap.

2. Methods

2.1. Dataset and preprocessing

The analysis used pretreatment EEG from 21 adults with treatment-resistant depression enrolled in a fully remote, multisite, double-blind, randomized sham-controlled home-based tDCS trial for major depressive disorder [8]. The analyzed cohort contained 7 remission cases and 14 non-remission cases in the patient-level outcome labels, with 18 female participants and a mean age of 37.1 ± 9.7 years. Table 1 summarizes the cohort, acquisition, and sweep configuration.

Baseline recordings were collected during eyes-closed rest with a 4-channel montage (`AF7`, `AF8`, `TP9`, `TP10`) sampled at 256 Hz. The available study exports comprised 10 s segments with 2,560 samples per channel, and the underlying recording sessions were stored either as a single 10 min file or as two 5 min files depending on participant. Across the cohort, the export set contained 1,203 windows, including 393 remission-labeled and 810 non-remission-labeled windows. This acquisition setup uses a limited frontal-temporal montage with clear portable-deployment relevance [10, 14]. The montage also constrained the downstream feature space toward frontal, temporal, cross-hemispheric, and frontal-temporal interaction measures.

For the present sweep, the processing configuration applied participant-level per-channel demeaning before upsampling, followed by a fourth-order Butterworth low-pass filter at 60 Hz, a downsampling step back to 256 Hz, and zero-overlap window slicing. The sweep then compared non-overlapping analysis windows of 2, 4, 6, 8, and 10 s, so the model inputs included both the native 10 s exports and shorter subdivisions derived from the same recordings. These choices materially affect downstream EEG feature distributions, especially in low-density or portable pipelines [11, 13, 47]. They also intersect with standardization and signal-quality concerns in wearable EEG settings [12, 15]. The methodological contribution of the present sweep lies in combining this broad handcrafted pool with participant-level outer validation, fold-internal feature selection, fold-internal class rebalancing, and post hoc recurrence summaries, rather than in introducing a single novel preprocessing transform [21, 33, 34].

2.2. Feature extraction

Feature extraction was performed on each preprocessed window. With all four channels available, the extractor generated 247 EEG-derived measures per window spanning channel-local descriptors, regional asymmetry terms, frontal-temporal interaction terms, synchrony summaries, and a small set of TP10-specific derived measures. Each channel contributed 39 local measures, comprising 21 spectral descriptors, 12 time-domain and Hjorth descriptors, and 6 entropy or nonlinear complexity descriptors, for 156 channel-local measures overall. The remaining 91 measures summarized asymmetry, frontal-temporal balance, cross-channel synchrony, and TP10-focused derived descriptors. This broad feature basis was deliberate. The goal of the study was to compare feature families under a leakage-safe sparse-selection regime rather than to pre-commit the analysis to a single biomarker hypothesis [34, 48]. Table 2 summarizes the extracted feature families and their computational rationale.

Table 1
Cohort, acquisition, and sweep configuration summary. Trial descriptors derive from the parent home-based tDCS study [8]; acquisition totals and sweep settings derive from the analyzed EEG exports and current experiment configuration.

Characteristic	Value
Parent trial and cohort descriptors	
Parent trial	Fully remote, multisite, sham-controlled home-based tDCS trial for MDD [8]
Participants	21
Outcome groups	7 remission, 14 non-remission
Sex (female/male)	18/3
Age (years, mean ± SD)	37.1 ± 9.7
EEG acquisition and export descriptors	
Rest condition	Eyes-closed resting state
EEG montage	4 channels (AF7, AF8, TP9, TP10)
Channel coverage	Frontal (AF7, AF8), temporal (TP9, TP10)
Sampling rate	256 Hz
Recording structure	10 min total; single file or two 5 min files
Available export segment length	10 s (2,560 samples/channel)
Total exported windows	1,203
Remission windows	393 (32.7%)
Non-remission windows	810 (67.3%)
Class ratio	2.06:1 (non-remission:remission)
Current sweep methodology settings	
Sweep window sizes	2, 4, 6, 8, 10 s
Inner-k sweep	1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70
Feature selector	SelectKBest (f-classif)
Training adjustments	LOPO-group equalization + SMOTE
Validation	Leave-one-participant-out
Evaluation level	Patient-level aggregation

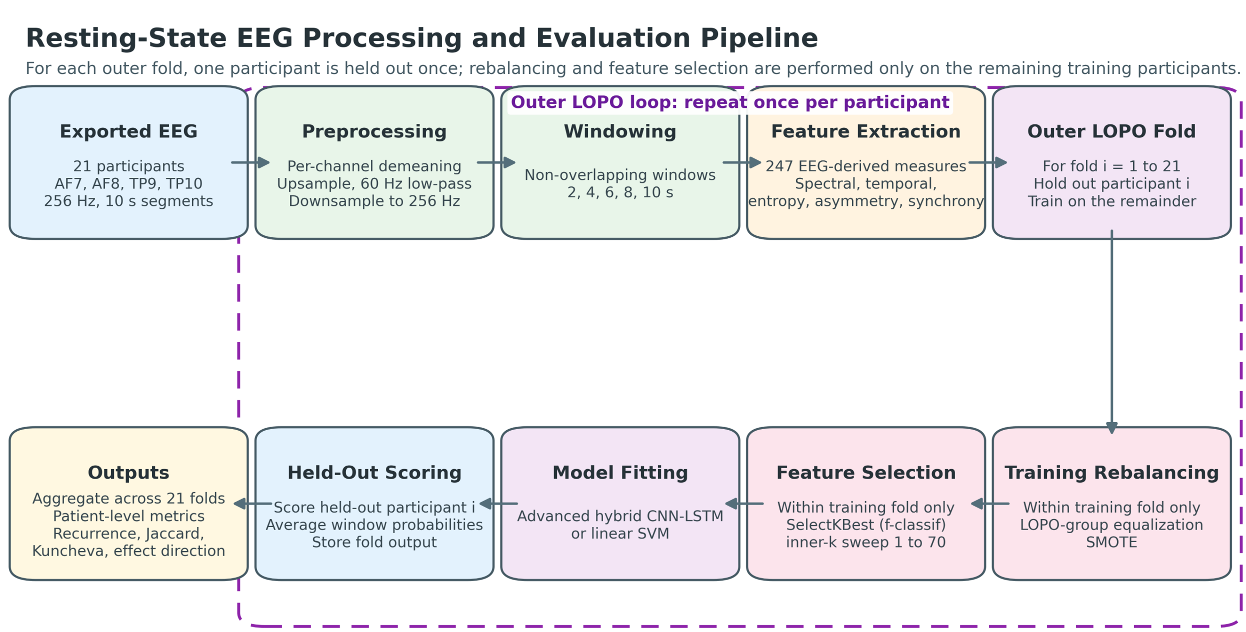

Figure 1: End-to-end analysis flow used in the study. For each outer leave-one-participant-out fold, one participant was held out once, and participant-group equalization, SMOTE, and feature selection were all applied only within the remaining training participants before held-out scoring and biomarker summarization.

Table 2
Summary of extracted EEG feature families per analysis window.

Feature family	Measures	Count	Why included
Channel-local spectral	Absolute and relative delta, theta, alpha, beta, and gamma power; total power; frequency-weighted power; SEF90 and SEF95; PSD mean, standard deviation, maximum, minimum, median, skewness, and kurtosis	21/channel (84 total)	Canonical resting-state EEG descriptors with clear band-level interpretation and strong precedent in depression EEG
Channel-local time domain and Hjorth	Mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, mean absolute deviation, Hjorth activity, mobility, and complexity	12/channel (48 total)	Captures amplitude dispersion, oscillatory rate, roughness, and short-window morphology with low computational cost
Channel-local entropy and nonlinear complexity	Sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis	6/channel (24 total)	Captures irregularity and complexity beyond band-power summaries
Band-power asymmetry and frontal-temporal interactions	Frontal and temporal asymmetry, hemispheric asymmetry, ipsilateral frontal-temporal ratios and normalized differences, and diagonal AF7-TP10 and AF8-TP9 interactions across frequency bands	70 total	Tests laterality and frontal-temporal balance within the four-channel montage
Synchrony and coupling	Pearson correlation and maximum normalized cross-correlation over six anatomical channel pairs, plus legacy cross-correlation summaries	18 total	Summarizes short-window coupling and cross-channel coordination
TP10-specific derived descriptors	Peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term	3 total	Targets concentrated temporal spectral structure with straightforward interpretation
Total EEG-derived features	Combined feature space per window	247 total	Broad hypothesis space for leakage-safe sparse selection

For each of AF7, AF8, TP9, and TP10, we computed 39 channel-local features. Spectral features were estimated from Welch power spectral densities [49] and included absolute band power in the delta, theta, alpha, beta, and gamma ranges, total power, relative band power normalized by total power, frequency-weighted power, spectral edge frequencies at 90% and 95%, and PSD summary statistics (mean, standard deviation, maximum, minimum, median, skewness, kurtosis). These measures were included because spectral content remains one of the most widely used and interpretable feature families in depression EEG work [16, 47, 50]. Temporal-channel spectral content has also appeared repeatedly in pretreatment response studies [51].

Time-domain descriptors were also computed for each channel. These included mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, and mean absolute deviation, together with Hjorth activity, mobility, and complexity [48, 52]. These features summarize amplitude dispersion, local oscillatory rate, signal roughness, and short-window morphology. They are computationally light and well suited to low-montage resting-state EEG, where clinically useful features must remain interpretable and robust under short windows.

Entropy and nonlinear complexity measures were computed per channel to capture irregularity beyond simple power summaries. The extractor included sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis. Complexity measures were calculated on a demeaned, standardized version of each signal, with long windows subsampled for numerical stability. These descriptors were retained because depression EEG studies have repeatedly implicated entropy, variability, and broader complexity structure [30, 53, 54]. That pattern also appears in pretreatment response and large-scale resting-state evaluations, even when the exact direction of effect depends on cohort, montage, and temporal scale [32, 51, 55].

The four-channel montage also supports a range of regional interaction features derived from cached band-power estimates. Traditional frontal asymmetry (AF7/AF8) and temporal asymmetry (TP9/TP10) were computed for each frequency band in two forms: a log-ratio index,

$$\mathrm{AI} = \log\left(\frac{P_R}{P_L}\right),$$

and a normalized difference,

$$\mathrm{ND} = \frac{P_R - P_L}{P_R + P_L}.$$

Using the same band-power cache, we also computed hemispheric asymmetry by averaging AF7+TP9 and AF8+TP10 within each hemisphere, ipsilateral frontal-temporal ratios and normalized differences on the left and right hemispheres, and diagonal frontal-temporal interactions for AF7-TP10 and AF8-TP9. These measures were included because reduced-channel depression EEG studies repeatedly point to laterality, temporal involvement, and frontal-temporal balance rather than to a single isolated electrode [31, 47, 56].

Pairwise synchrony features were added for six anatomically meaningful channel pairs: `AF7-AF8`, `TP9-TP10`, `AF7-TP9`, `AF8-TP10`, `AF7-TP10`, and `AF8-TP9`. For each pair we computed maximum normalized cross-correlation and Pearson correlation, and the extractor also retained a legacy set of raw cross-correlation summaries across all channel pairs. These features were intended to capture short-window synchrony and coupling patterns that have been emphasized in resting-state depression EEG and neuromodulation response work [31, 32, 44].

Finally, the extractor included three exploratory TP10-specific derived descriptors: peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term. These summarize whether the `TP10` spectrum contains a sharp dominant peak, how concentrated that peak is relative to total power, and whether that concentration covaries with ipsilateral frontal-temporal beta balance. They were included because temporal-channel spectral content has recurrently appeared in low-montage depression EEG analyses, while still remaining interpretable enough to survive foldwise sparse selection if informative [47, 50].

2.3. Sweep design and validation

The experiment sweep compared two model families over the same feature-selection protocol: an advanced hybrid 1D CNN-LSTM and a linear SVM. All runs used sequential window ordering, SelectKBest with ANOVA F statistics, leave-one-participant-out outer evaluation, leave-one-participant-out training-group equalization, and SMOTE within training folds [57]. The sweep varied the per-fold feature-selection budget (`inner-k`) across 15 values from 1 to 70 and repeated the search across 5 window sizes, yielding 150 successful runs in total (75 per model).

This design matters for interpretation. The outer evaluation remained subject independent throughout, so no participant contributed both training and test windows to the same outer fold. Feature selection, participant-group equalization, and synthetic oversampling were all restricted to the outer training folds, which is important because leakage at these steps can materially inflate reported biomarker performance [33]. The `inner-k` parameter controlled how many features were selected inside each outer training fold. The feature-selection stability panels in the study describe shared pipeline behavior because both classifiers consumed the same selected features for matched sweep settings.

2.4. Models

Both classifiers operated on the fold-selected handcrafted feature vectors produced by the current `inner-k` setting. Patient-level outputs were derived by averaging held-out window probabilities within each participant before computing ROC-AUC and the other clinical metrics. Table 3 summarizes the model architecture and training setup used in the sweep.

The SVM baseline used a linear kernel with `C=1.0` and probability output enabled. When SMOTE was active inside the training folds, class weighting was not applied in parallel, so synthetic oversampling and explicit class weighting were not used simultaneously.

The hybrid model treated the selected feature vector as a learned one-dimensional representation before recurrent modeling. In the best-performing 6 s, `inner-k=1` run, the input features were first projected to a 128-dimensional embedding with normalization and dropout, then mapped into a 64-position latent representation for three convolutional blocks. These blocks used channel progressions of `[64,64]`, `[128,128]`, and `[256,128]` with kernel sizes `[5,3]`, `[3,3]`, and `[3,1]`, together with batch normalization, max pooling, and dropout after each stage.

Sequence modeling then used two bidirectional LSTM layers with 128 and 64 hidden units, followed by eight-head self-attention with residual layer normalization and dropout. The recurrent summary was globally pooled and concatenated with pooled multi-scale CNN outputs through feature-pyramid fusion before two dense blocks of 256 and 128 units and a final two-logit output layer. Training used AdamW, cosine warm restarts, label smoothing, gradient clipping, mixed precision, and early stopping with best-state restore.

The architectural choices were intended to distribute the modeling burden across complementary stages. The projection layer compressed the sparse selected features into a stable latent representation before deeper processing. The convolutional front end summarized short-range interactions and produced progressively richer multi-scale feature maps. The bidirectional LSTM layers then provided recurrent processing after the convolutional front end, while self-attention allowed the model to reweight informative positions after recurrent encoding. Feature-pyramid fusion retained both shallower and deeper convolutional summaries, which is useful when the most discriminative structure may not reside at a single representational depth. The dense head provided a final nonlinear compression stage before binary classification. Batch normalization, dropout, label smoothing, gradient clipping, and early stopping were included to stabilize optimization and limit overfitting in a relatively small cohort.

2.5. Outcome metrics and biomarker summaries

The primary endpoint was patient-level ROC-AUC, computed from participant probabilities aggregated over held-out windows. Secondary patient-level metrics included PR-AUC, balanced accuracy, accuracy, F1, MCC, precision, recall, and specificity. The pipeline also reported bootstrap confidence intervals and a permutation test for the patient-level ROC-AUC summary. All main study claims use patient-level metrics. Window-level scores appear only as intermediate outputs.

To characterize biomarker behavior across outer folds, we used several post hoc summaries: mean pairwise Jaccard overlap, Kuncheva index, unique feature count,

Table 3
Model architecture and training summary for the two evaluated classifier families.

Component	Configuration
Input to both models	Fold-selected handcrafted feature vector from the current `inner-k` setting. Patient-level predictions were computed by averaging held-out window probabilities within each participant.
SVM baseline	Linear-kernel SVM with `C=1.0` and probability estimates enabled. When SMOTE was active in the training folds, class weighting was disabled to avoid double balancing.
Hybrid input embedding	Linear projection from the selected feature vector to 128 latent units, followed by normalization, a nonlinear activation, dropout 0.1, and remapping to a learned one-dimensional sequence of length 64.
CNN trunk	Three 1D convolutional blocks with filter progressions `[64,64]`, `[128,128]`, and `[256,128]`; kernel sizes `[5,3]`, `[3,3]`, and `[3,1]`; max-pooling factors 2, 2, and 1; batch normalization; and dropout `[0.1, 0.2, 0.3]`.
Recurrent stage	Two bidirectional LSTM layers with 128 and 64 hidden units followed the CNN trunk.
Attention and fusion	Eight-head self-attention with dropout 0.1 and residual layer normalization, followed by global mean pooling. Multi-scale pooled CNN features were concatenated through feature-pyramid fusion.
Dense head	Two dense blocks of 256 and 128 units with batch normalization, nonlinear activations, and dropout 0.3 and 0.4, followed by a two-logit output layer.
Optimization	AdamW with learning rate 0.001 and weight decay 0.01; cosine warm restarts to a minimum learning rate of 10^{-6}; batch size 8192 before fold-size adjustment; label smoothing 0.05; gradient clipping at 1.0; mixed precision; maximum 100 epochs; early stopping patience 5 with best-state restore.

top-feature share, class-conditional selection deltas, and effect-direction consistency. These summaries are descriptive. They characterize recurrence and directionality across folds and do not support causal interpretation. Low overlap can coexist with recurring features that retain consistent class-separation direction when selected [12, 16, 17].

3. Results

3.1. Sweep-level comparison

Across the 150 successful runs, the hybrid model outperformed the SVM baseline on the main patient-level discrimination metrics over most of the sweep landscape. The performance advantage was visible when averaging over window sizes (Figure 2) and when averaging over `inner-k` settings (Figure 3). The best individual hybrid run occurred at 6 s windows with `inner-k=1`, whereas the best SVM run occurred at 8 s windows with `inner-k=30`.

Figure 3 shows that the relationship between ROC-AUC and `inner-k` differed across the two model families in the low-budget region. The hybrid model retained clearly stronger ROC-AUC across the sparsest budgets and achieved the highest individual ROC-AUC in the entire sweep at `inner-k=1`. The linear SVM did not show the same pattern and reached its best ROC-AUC only at the broader `inner-k=30` setting. Sparse per-fold selection therefore aligned with peak discrimination only for the more expressive model.

Figures 4 and 5 show a second related pattern. Increasing `inner-k` increased both hard-set overlap and the total number of unique selected features. Larger feature budgets did not produce the strongest patient-level discrimination. The sweep concentrated its best hybrid performance at the sparsest end of the per-fold selection range. This pattern motivates the paper's focus on sparse

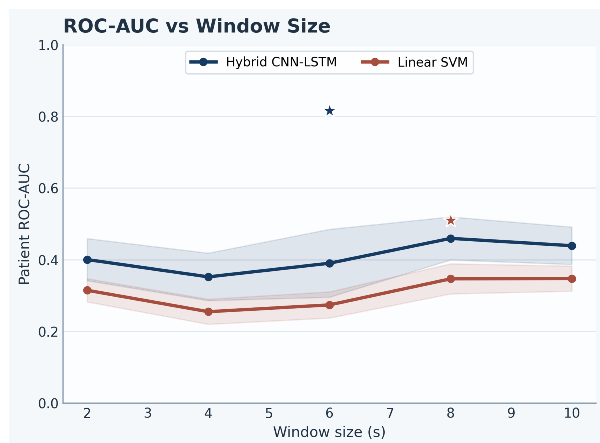

Figure 2: Patient-level ROC-AUC by window size across the sweep. Stars mark the best individual hybrid and SVM runs.

biomarker discovery and suggests that model capacity and per-fold sparsity mattered together rather than independently.

3.2. Best-run patient-level performance

The best hybrid configuration achieved patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, F1 0.714, and MCC 0.571 over 21 held-out participants (Figures 6 and 7; Table 4). Its ROC-AUC bootstrap interval ranged from 0.600 to 0.990, and the permutation test remained significant ($p = 0.008$), which is notable given the relatively small cohort.

The corresponding confusion matrix contained 12 true negatives, 5 true positives, 2 false positives, and 2 false negatives (Figure 8). The metric summary in Figure 9 shows the same run in condensed form.

The best SVM baseline reached balanced accuracy 0.643 and ROC-AUC 0.510, with a non-significant permutation

Table 4
Best-performing configuration for each model family.

Model	Window	Inner-k	ROC-AUC	PR-AUC	Bal. Acc.	F1	MCC	Feat./fold	Unique feat.	Mean Jaccard
Hybrid CNN-LSTM	6	1	0.816	0.579	0.786	0.714	0.571	1.0	7	0.190
Linear SVM	8	30	0.510	0.411	0.643	0.533	0.277	30.0	103	0.360

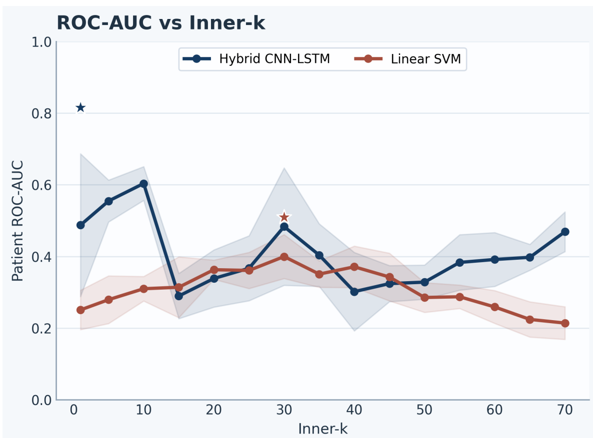

Figure 3: Patient-level ROC-AUC by `inner-k` across the sweep. The hybrid model retained the strongest discrimination in the sparsest budget region.

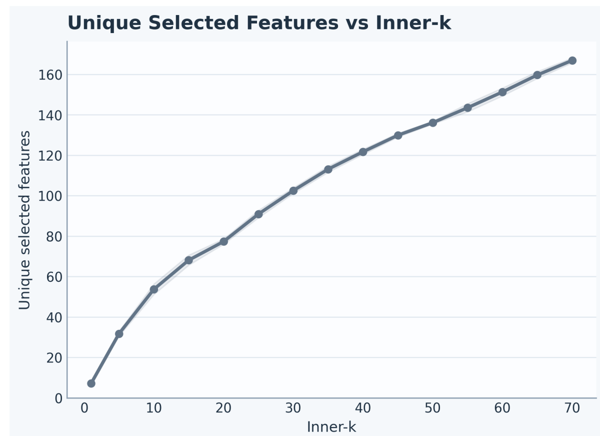

Figure 5: Unique selected features by `inner-k`. Larger per-fold budgets produced markedly larger discovered feature pools.

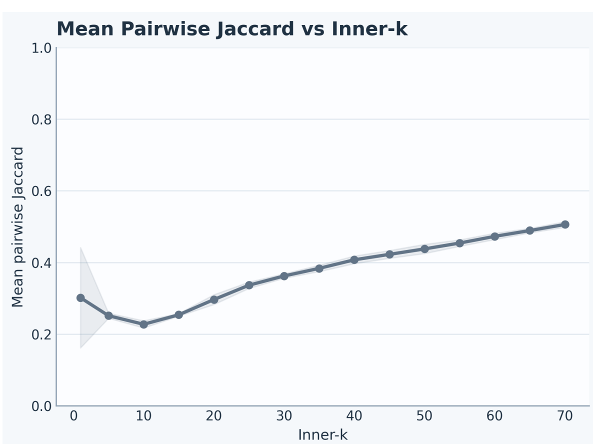

Figure 4: Mean pairwise Jaccard overlap by `inner-k`. Hard-set overlap increased with larger per-fold feature budgets.

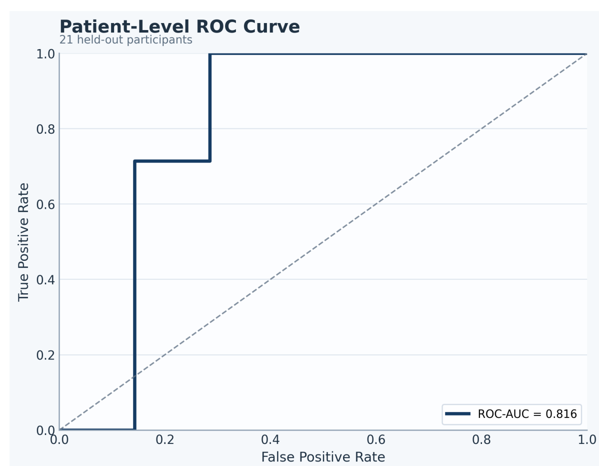

Figure 6: Patient-level ROC curve for the best hybrid run.

result (p = 0.469). This weaker performance came with a much larger per-fold feature budget (`inner-k=30`) and a far larger overall candidate set. The contrast indicates that the sparse hybrid pipeline extracted stronger patient-level discrimination from a much smaller subset.

3.3. Sparse biomarker set with modest hard-set overlap

The winning hybrid run selected an average of 1 feature per fold and yielded only 7 unique recurrent features across the entire outer-validation process (Table 5). That degree of sparsity is one of the most distinctive findings in the sweep. Hard-set overlap was modest: mean pairwise Jaccard 0.190 and mean Kuncheva index 0.187. Figures 10 and 11 summarize the small recurrent pool together with the modest chance-adjusted stability summary.

A small recurring feature set emerged repeatedly, though not in every fold, and the selected set was dominated by temporal and frontal-temporal descriptors. The most recurrent feature was `tp10_spectral_entropy`, selected in 8 of 21 folds. The next most recurrent features were `left_frontal_temporal_diff_gamma`, `left_frontal_temporal_diff_beta`, and `af7_zero_crossings`.

Table 5
Recurrent biomarkers from the best hybrid run. Δ selection is remission minus non-remission selection share.

Feature	Folds	Share	Δ sel.	Sign cons.	Mean d	Effect direction
tp10_spectral_entropy	8	0.381	-0.571	1.000	-0.935	Non-remission > remission
left_frontal_temporal_diff_gamma	4	0.190	-0.071	1.000	-1.059	Non-remission > remission
left_frontal_temporal_diff_beta	3	0.143	0.000	1.000	-0.915	Non-remission > remission
af7_zero_crossings	3	0.143	0.429	1.000	0.814	Remission > non-remission
tp10_hurst	1	0.048	0.143	1.000	-1.010	Non-remission > remission
af7_tp10_diff_gamma	1	0.048	0.143	1.000	-1.203	Non-remission > remission
frontal_asymmetry_ratio_delta	1	0.048	-0.071	1.000	0.794	Remission > non-remission

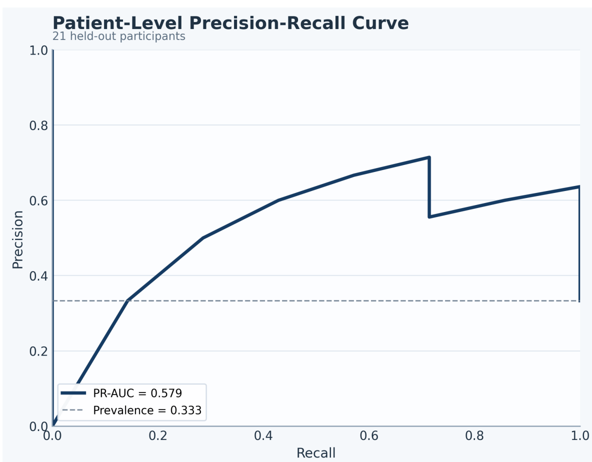

Figure 7: Patient-level precision-recall curve for the best hybrid run.

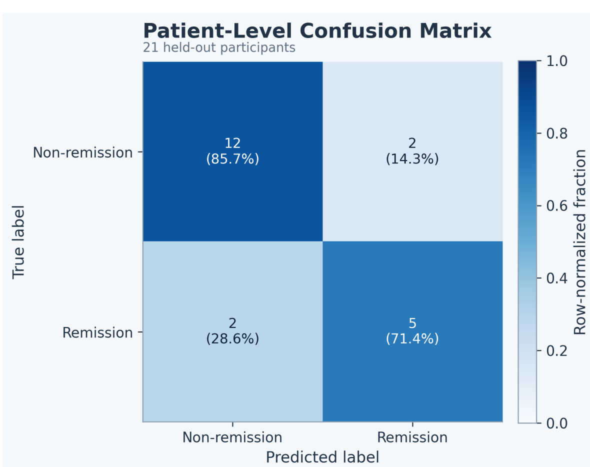

Figure 8: Patient-level confusion matrix for the best hybrid run.

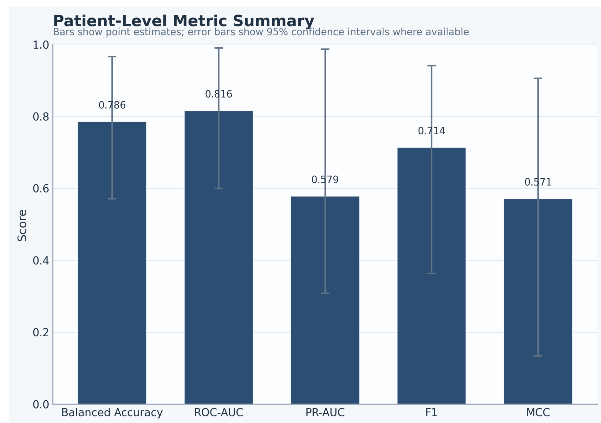

Figure 9: Patient-level metric summary for the best hybrid run.

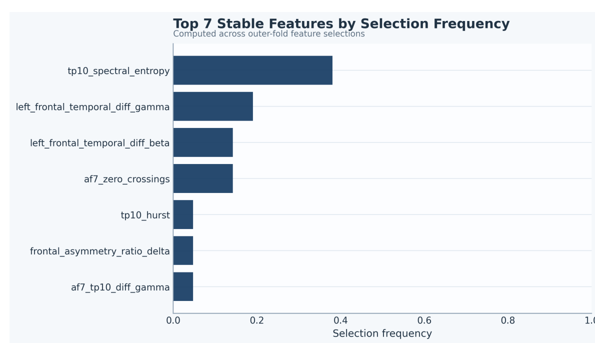

Figure 10: Selection-frequency ranking for the recurrent biomarkers from the best hybrid run.

3.4. Effect-direction consistency and class-conditional patterns

The recurrent hybrid biomarkers showed perfect mean sign consistency across the effect-stability summary. This supports treating them as promising biomarkers with stable directionality across the folds in which they were selected (Figure 12).

`tp10_spectral_entropy` and the frontal-temporal gamma and beta difference terms were repeatedly associated with the non-remission group, whereas `af7_zero_crossings` was more frequently selected in remission folds. The class-conditional selection frequencies make that separation visible at the level of foldwise recurrence (Figure 13).

The effect-size summaries followed the same broad pattern, with `af7_zero_crossings` retaining a positive mean effect size and the main TP10 and frontal-temporal features retaining negative mean effects (Figure 14). This distinction matters clinically. The selection-frequency deltas do not justify definitive mechanistic claims. They provide a concrete interpretation target for follow-up validation. The

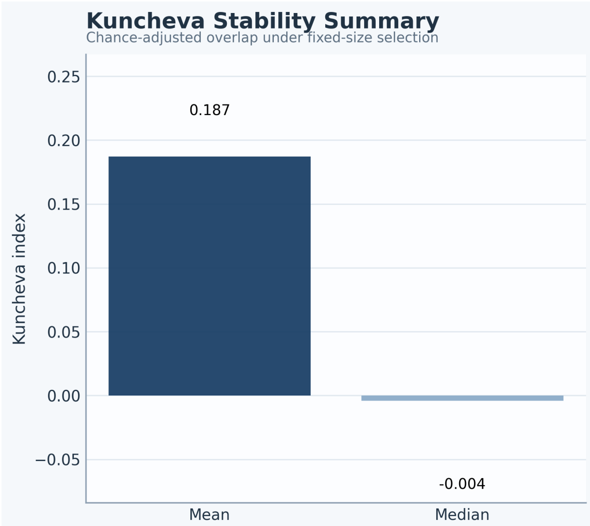

Figure 11: Kuncheva stability summary for the recurrent biomarkers from the best hybrid run.

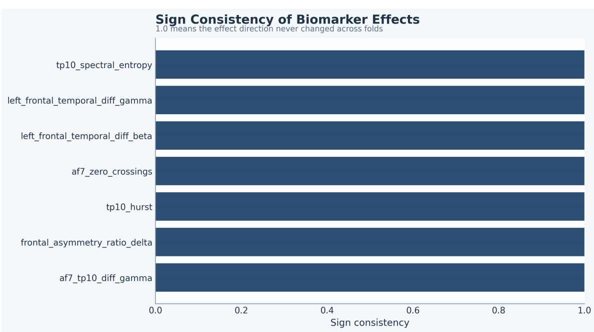

Figure 12: Sign-consistency summary for the recurrent biomarkers from the best hybrid run. All recurrent features retained the same effect direction across the folds in which they were selected.

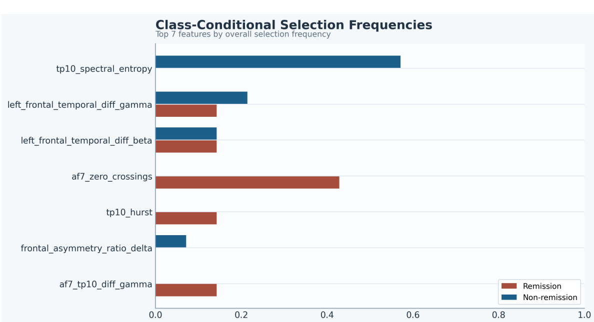

Figure 13: Class-conditional selection frequencies for the recurrent biomarkers from the best hybrid run. Remission is shown in red and non-remission in blue.

best hybrid pipeline contributes two linked observations: useful patient-level discrimination can be achieved with an extremely sparse feature budget, and the resulting recurrent biomarkers show interpretable directionality even when fold-to-fold set overlap remains modest.

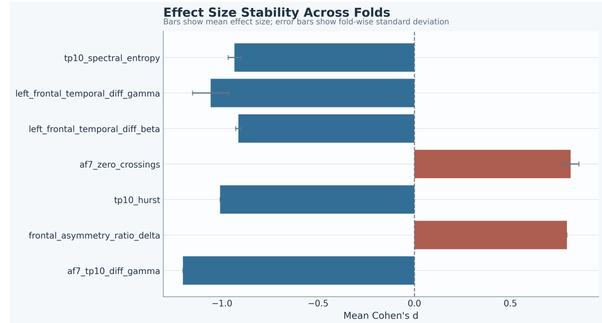

Figure 14: Effect-size stability for the recurrent biomarkers from the best hybrid run. Positive values indicate remission-skewed effects and negative values indicate non-remission-skewed effects.

4. Discussion

The strongest patient-level discrimination in this sweep emerged from the sparsest tested configuration. The best hybrid run used `inner-k=1`, selected only 7 unique recurrent features overall, and exceeded the best linear baseline by a large margin on ROC-AUC and MCC. Figure 3 also shows that this was not a generic effect of smaller feature budgets across both models. In the low-budget region, the hybrid model retained stronger ROC-AUC than the linear SVM and reached its peak performance at the sparsest tested setting, whereas the linear model reached its best ROC-AUC only after a broader per-fold budget. In this cohort, the value of the more expressive model and the value of strong sparsity appear to have operated together.

One plausible explanation is that aggressive per-fold pruning removed weak or redundant handcrafted features and exposed a compact signal set whose interactions were more useful to the hybrid architecture than to a linear decision boundary. With only a few retained descriptors, the convolutional, recurrent, and attention stages may have been better positioned to combine complementary temporal and frontal-temporal information without dilution from a larger pool of unstable candidates. The linear baseline appears to have depended more on accumulating a broader set of moderate additive cues before reaching its best ROC-AUC. This interpretation remains provisional, but it is consistent with the sweep pattern and with the marked separation between the best sparse hybrid run and the broader-budget linear baseline.

This result fits part of the EEG depression literature and supports a measured interpretation. Prior work has argued that EEG can provide clinically useful baseline markers in major depressive disorder and treatment-response settings [16–18]. Related discussions have also emphasized tighter biomarker discipline, better standardization, and more realistic expectations about generalization [12, 20, 58]. Machine learning studies have likewise reported encouraging treatment-response signals, including combinations of EEG and clinical covariates [24–26]. Other modelling studies reach similar conclusions while still exposing limits from data heterogeneity, cohort size, or insufficient interpretability [27–29]. Recent predictive overviews and reviews continue to highlight the same translational tension [22, 23].

The present sweep adds a feature-selection view to that literature. Figures 4 and 5 show that larger feature budgets yield more overlap and many more unique discovered features. Those larger budgets do not improve discrimination. The panels support a small recurring biomarker set with consistent direction of class separation. That reading keeps the modest Jaccard and Kuncheva scores in view and gives appropriate weight to the stronger effect-direction signal. Methodologically, this framing matters because the depression EEG biomarker literature still contains substantial publication bias, limited out-of-sample validation, and known vulnerability to leakage from pre-cross-validation feature processing [21, 33]. The present pipeline addresses that problem by embedding selection and rebalancing inside participant-level outer folds and then summarizing recurrence and directionality instead of reporting only a single global feature list [34].

The same clinical program has also supported other signal representations in recent PSD-based and PLV-based studies, which located informative response-related signal in low-density temporal and frontal-temporal patterns [45, 46]. In the present sweep, the recurrent feature set again concentrated in TP10-derived and frontal-temporal descriptors. That regional convergence supports the view that the low-montage recordings contain response-relevant information across multiple feature families.

The specific biomarkers in the present sweep are plausible enough to deserve follow-up. `tp10_spectral_entropy` was the most frequently selected feature and skewed toward non-remission-associated folds. Entropy and related nonlinear descriptors have appeared repeatedly in depression EEG studies [30, 53, 54]. Pretreatment multiscale entropy and signal-variability measures have also been linked to antidepressant treatment response, although the reported direction depends on temporal scale and recording context [51]. That nuance is important because a recent large resting-state evaluation found complexity metrics to be stable but not to support a simple uniform depression-complexity association [55]. The present TP10 entropy finding therefore fits a broader complexity literature and supports a credible temporal biomarker signal in this cohort, while still needing external confirmation.

The frontal-temporal beta and gamma difference features also have prior support at the family and regional level. Temporal electrodes, higher-frequency band features, and paired asymmetry terms have been reported as informative in depression EEG studies, and pretreatment frontal-temporal features have been linked to treatment outcome in MDD [47, 50]. The current beta and gamma difference terms are therefore better read as family-level extensions of that regional pattern than as exact replications of a previously fixed biomarker.

`af7_zero_crossings` had the clearest remission-skewed selection pattern in the current sweep, but it has less direct precedent as a named biomarker. AF7-centered signal relevance has appeared in adjacent low-montage work [59], and broader frontal EEG literature continues to implicate asymmetry and signal-complexity families in depression [55, 56]. The specific zero-crossing formulation used here is therefore best regarded as a plausible frontal biomarker in this setting and should be tested in independent cohorts. Taken together, the recurrent set emphasized temporal and frontal-temporal descriptors, with limited concentration in any single frontal asymmetry family. This regional profile is compatible with the broader low-montage findings discussed above. The current study contributes a sparse handcrafted biomarker signature with explicit recurrence and effect-direction summaries.

Related tDCS studies in bipolar depression from the same group point in a similar low-montage direction. PSD-based modeling again highlighted AF7 and TP10 signal content, and PLV analyses emphasized frontal-temporal and temporoparietal synchronization patterns [59, 60]. These bipolar studies address a different diagnosis and treatment protocol. They support reduced-montage feasibility and signal plausibility in adjacent affective cohorts.

The dataset is relatively small ($n = 21$), the confidence intervals are wide, and there is no external validation set. The bottom-row sweep panels describe the selection pipeline itself because the same feature-selection layer feeds both classifiers. Class-conditional ratios also become unstable when a feature is absent from one class, so the study uses deltas and effect direction as the main descriptive summaries. The recurring features can be treated as plausible biomarkers from this cohort that now warrant follow-up validation.

Future work should test whether the same sparse feature set recurs under independent cohorts, alternative low-density montages, and explicit external validation [40]. Adjacent response-prediction work in depression and neuromodulation provides useful targets for that next stage [35, 36, 38]. Home-based and transport-oriented studies offer one clear direction for follow-up [61–63]. More recent response-signature studies also provide relevant benchmarks for that next phase [41–43]. Related protocol and review work in treatment-resistant depression can help frame that validation effort [44, 64]. These extensions should keep stringent patient-level evaluation and sparse interpretable feature budgets in the core study design.

5. Conclusion

In this sweep, low-montage resting-state EEG supported useful patient-level discrimination of tDCS response in adults with treatment-resistant depression, and the strongest result came from an extremely sparse feature-selection regime. The best hybrid run achieved ROC-AUC 0.816 while selecting only 1 feature per fold and 7 unique recurrent features overall. Hard-set overlap across folds remained modest. The present evidence supports a small, interpretable biomarker signature with consistent effect direction under strict participant-level validation.

That combination of parsimony, interpretability, and subject-independent evaluation defines the main contribution of the present draft. External validation in larger cohorts is the next requirement. The current results support the feasibility of sparse short-timescale EEG biomarker discovery for tDCS response in a clinically realistic low-density setting.

References

[1] Roumen Milev, Peter Giacobbe, Sidney H. Kennedy, Daniel M. Blumberger, Zafiris J. Daskalakis, Jonathan Downar, Mandana Modirrousta, Simon Patry, Fidel Vila Rodriguez, Raymond W. Lam, Glenda MacQueen, Sagar V. Parikh, Arun Ravindran, and the CANMAT Depression Work Group. Canadian network for mood and anxiety treatments (canmat) 2016 clinical guidelines for the management of adults with major depressive disorder. *The Canadian Journal of Psychiatry*, 2016. doi: 10.1177/0706743716660033. URL https://doi.org/10.1177/0706743716660033.

[2] Julian Mutz, Daniel R Edgcumbe, Andre R Brunoni, and Cynthia HY Fu. Efficacy and acceptability of non-invasive brain stimulation for the treatment of adult unipolar and bipolar depression: a systematic review and meta-analysis of randomised sham-controlled trials. *Neuroscience & Biobehavioral Reviews*, 92:291–303, 2018.

[3] Lais B Razza, Priscila Palumbo, Adriano H Moffa, Andre F Carvalho, Marco Solmi, Colleen K Loo, and Andre Russowsky Brunoni. A systematic review and meta-analysis on the effects of transcranial direct current stimulation in depressive episodes. *Depression and Anxiety*, 37(7):594–608, 2020.

[4] Paulo S. Boggio, Sergio P. Rigonatti, Rafael Bernardon Ribeiro, Martin Luiz Myczkowski, Michael A. Nitsche, Alvaro Pascual Leone, and Felipe Fregni. A randomized, double-blind clinical trial on the efficacy of cortical direct current stimulation for the treatment of major depression. *The International Journal of Neuropsychopharmacology*, 2007. doi: 10.1017/s1461145707007833. URL https://doi.org/10.1017/s1461145707007833.

[5] André R Brunoni, Adriano H Moffa, Felipe Fregni, Ulrich Palm, Frank Padberg, Daniel M Blumberger, Zafiris J Daskalakis, Djamila Bennabi, Emmanuel Haffen, Angelo Alonzo, et al. Transcranial direct current stimulation for acute major depressive episodes: meta-analysis of individual patient data. *The British Journal of Psychiatry*, 208(6): 522–531, 2016.

[6] Hayley Thair, Amy L. Holloway, Roger Newport, and Alastair D. Smith. Transcranial direct current stimulation (tdcs): A beginner's guide for design and implementation. *Frontiers in Neuroscience*, 2017. doi: 10.3389/fnins.2017.00641. URL https://doi.org/10.3389/fnins.2017.00641.

[7] Steven C. Cramer, Mriganka Sur, Bruce H. Dobkin, Charles P. OBrien, Terence D. Sanger, John Q. Trojanowski, Judith M. Rumsey, Ramona Hicks, Judy L. Cameron, David Chen, Wen Chen, Leonardo G. Cohen, C. deCharms, C. J. Duffy, Guinevere F. Eden, E. E. Fetz, Rosemarie Filart, M C Freund, Steven Grant, Suzanne N. Haber, Peter W. Kalivas, Bryan Kolb, Arthur F. Kramer, Michael P. Lynch, Helen S. Mayberg, Patrick S. McQuillen, Ralph Nitkin, A. Pascual Leone, Patricia A. Reuter Lorenz, Nicholas D. Schiff, Anu Sharma, L. Shekim, Michael P. Stryker, Edith V. Sullivan, and Sophia Vinogradov. Harnessing neuroplasticity for clinical applications. *Brain*, 2011. doi: 10.1093/brain/awr039. URL https://doi.org/10.1093/brain/awr039.

[8] Rachel D Woodham, Sudhakar Selvaraj, Nahed Lajmi, Harriet Hobday, Gabrielle Sheehan, Ali-Reza Ghazi-Noori, Peter J Lagerberg, Maheen Rizvi, Sarah S Kwon, Paulette Orhii, et al. Home-based transcranial direct current stimulation treatment for major depressive disorder: a fully remote phase 2 randomized sham-controlled trial. *Nature medicine*, 31(1):87–95, 2025.

[9] Rachael M Rimmer, Rachel D Woodham, Sharon Cahill, and Cynthia HY Fu. Acceptability of home-based transcranial direct current stimulation (tdcs) in major depression: a qualitative analysis of individual experiences. *Mental Health Review Journal*, 29(1):79–91, 2024.

[10] Alexander J Casson, David C Yates, Shelagh JM Smith, John S Duncan, and Esther Rodriguez-Villegas. Wearable electroencephalography. *IEEE engineering in medicine and biology magazine*, 29(3):44–56, 2010.

[11] Priya Miranda, Christopher D Cox, Michael Alexander, Slav Danev, and Jonathan RT Lakey. Overview of current diagnostic, prognostic, and therapeutic use of eeg and eeg-based markers of cognition, mental, and brain health. *Integr. Mol. Med*, 6(5), 2019.

[12] Leif Simmatis, Emma E. Russo, Joseph Geraci, Irene E. Harmsen, and Nardin Samuel. Technical and clinical considerations for electroencephalography-based biomarkers for major depressive disorder. *npj Mental Health Research*, 2023. doi: 10.1038/s44184-023-00038-7. URL https://doi.org/10.1038/s44184-023-00038-7.

[13] Siuly Siuly, Yan Li, and Yanchun Zhang. Electroencephalogram (eeg) and its background. In *EEG signal analysis and classification: Techniques and applications*, pages 3–21. Springer, 2017.

[14] Sally M Stoyell, Janina Wilmskoetter, Mary-Ann Dobrota, Dhinakaran M Chinappen, Leonardo Bonilha, Mark Mintz, Benjamin H Brinkmann, Susan T Herman, Jurriaan M Peters, Serge Vulliemoz, et al. High-density eeg in current clinical practice and opportunities for the future. *Journal of clinical neurophysiology*, 38(2):112–123, 2021.

[15] BABILONI, CLAUDIO, Boeijinga PH, Brunovsky M, Drinkenburg WH, Ffytche DH, Freeman J, Hegerl U, Hirata K, IPEG Pharmaco EEG Guidelines C.o.m.m.i.t.t.e.e. Collaborators, Jobert M, Kinoshita T, Knott VJ, Lopes Da Silva FH, Matousek M, Mucci A, Nottage JF, Olbrich S, Prichep LS, Ruigt GS, Saletu B, Stancak A, Strik WK, van Gerven JM, Wilson FJ, and Wise RG. Guidelines for the recording and evaluation of pharmaco-eeg data in man: the international pharmaco-eeg society (ipeg). 2012. doi: 10.1159/000343478. URL https://iris.uniroma1.it/bitstream/11573/802874/1/Jobert_Guidelines_2012.pdf.

[16] Sebastian Olbrich and Martijn Arns. Eeg biomarkers in major depressive disorder: Discriminative power and prediction of treatment response. *International Review of Psychiatry*, 2013. doi: 10.3109/09540261.2013.816269. URL https://doi.org/10.3109/09540261.2013.816269.

[17] Sebastian Olbrich, Rik van Dinteren, and Martijn Arns. Personalized medicine: Review and perspectives of promising baseline eeg biomarkers in major depressive disorder and attention deficit hyperactivity disorder. *Neuropsychobiology*, 2015. doi: 10.1159/000437435. URL https://doi.org/10.1159/000437435.

[18] Andrew F. Leuchter, Ian A. Cook, Steven P. Hamilton, Katherine L. Narr, Arthur Toga, Aimee M. Hunter, Kym Faull, Julian Whitelegge, Anne M. Andrews, Joseph Loo, Baldwin Way, Stanley F. Nelson, Steven Horvath, and Barry D. Lebowitz. Biomarkers to predict antidepressant response. *Current Psychiatry Reports*, 2010. doi: 10.1007/s11920-010-0160-4. URL https://pubmed.ncbi.nlm.nih.gov/20963521/.

[19] Cynthia HY Fu and Sergi G Costafreda. Neuroimaging-based biomarkers in psychiatry: clinical opportunities of a paradigm shift. *The Canadian Journal of Psychiatry*, 58(9):499–508, 2013.

[20] Cynthia HY Fu, Yong Fan, and Christos Davatzikos. Addressing heterogeneity (and homogeneity) in treatment mechanisms in depression and the potential to develop diagnostic and predictive biomarkers. *NeuroImage: Clinical*, 24:101997, 2019.

[21] Alik S. Widge, M. Taha Bilge, Rebecca Montana, Weilynn Chang, Carolyn I. Rodriguez, Thilo Deckersbach, Linda L. Carpenter, Ned H. Kalin, and Charles B. Nemeroff. Electroencephalographic biomarkers for treatment response prediction in major depressive illness: A meta-analysis. *American Journal of Psychiatry*, 176(1):44–56, 2019. doi: 10.1176/appi.ajp.2018.17121358.

[22] Devon Watts, Rafaela Fernandes Pulice, Jim Reilly, Andre R Brunoni, Flávio Kapczinski, and Ives Cavalcante Passos. Predicting treatment response using eeg in major depressive disorder: A machine-learning meta-analysis. *Translational psychiatry*, 12(1):332, 2022.

[23] SE Cohen, JB Zantvoord, BN Wezenberg, JG Daams, CLH Bockting, Damiaan Denys, and GA van Wingen. Electroencephalography for predicting antidepressant treatment success: a systematic review and meta-analysis. *Journal of Affective Disorders*, 321:201–207, 2023.

[24] Alaa M Al-Kaysi, Ahmed Al-Ani, Colleen K Loo, Tamara Y Powell, Donel M Martin, Michael Breakspear, and Tjeerd W Boonstra. Predicting tdcs treatment outcomes of patients with major depressive disorder using automated eeg classification. *Journal of affective disorders*, 208:597–603, 2017.

[25] Natalia Jaworska, Sara de la Salle, Mohamed Hamza Ibrahim, Pierre Blier, and Verner Knott. Leveraging machine learning approaches for predicting antidepressant treatment response using electroencephalography (eeg) and clinical data. *Frontiers in Psychiatry*, 2019. doi: 10.3389/fpsyt.2018.00768. URL https://doi.org/10.3389/fpsyt.2018.00768.

[26] Thomas Oakley, Jonathan Coskuner, Andrew Cadwallader, Maryam Ravan, and Gary Hasey. Eeg biomarkers to predict response to sertraline and placebo treatment in major depressive disorder. *IEEE Transactions on Biomedical Engineering*, 2022. doi: 10.1109/tbme.2022.3204861. URL https://doi.org/10.1109/tbme.2022.3204861.

[27] Reza Shalbaf, Colleen A. Brenner, C. Pang, Daniel M. Blumberger, Jonathan Downar, Zafiris J. Daskalakis, Joseph Tham, Raymond W. Lam, Faranak Farzan, and Fidel Vila-Rodriguez. Non-linear entropy analysis in eeg to predict treatment response to repetitive transcranial magnetic stimulation in depression. *Frontiers in Pharmacology*, 2018. doi: 10.3389/fphar.2018.01188. URL https://doi.org/10.3389/fphar.2018.01188.

[28] Elias Ebrahimzadeh, Mostafa Asgarinejad, Sarah Saliminia, Sarvenaz Ashoori, and Masoud Seraji. Predicting clinical response to transcranial magnetic stimulation in major depression using time-frequency eeg signal processing. *Biomedical Engineering Applications Basis and Communications*, 2021. doi: 10.4015/s1016237221500484. URL https://doi.org/10.4015/s1016237221500484.

[29] A. Zhdanov et al. Use of machine learning for predicting escitalopram treatment outcome from eeg in adults with depression. *Human Brain Mapping*, 2020. Placeholder citation for Zhdanov SVM study.

[30] Subha D Puthankattil and Paul K Joseph. Classification of eeg signals in normal and depression conditions by ann using rwe and signal entropy. *J. Mech. Med. Biol.*, 12(04):1240019, September 2012.

[31] S Ghiasi, C Dell'Acqua, S Messerotti Benvenuti, E P Scilingo, C Gentili, G Valenza, and A Greco. Classifying subclinical depression using EEG spectral and connectivity measures. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pages 2050–2053. IEEE, November 2021.

[32] Shanguang Zhao, Siew-Cheok Ng, Selina Khoo, and Aiping Chi. Temporal and spatial dynamics of EEG features in female college students with subclinical depression. *Int. J. Environ. Res. Public Health*, 19(3), February 2022.

[33] Miseon Shim, Seung-Hwan Lee, and Han-Jeong Hwang. Inflated prediction accuracy of neuropsychiatric biomarkers caused by data leakage in feature selection. *Scientific Reports*, 11(1), apr 2021. doi: 10.1038/s41598-021-87157-3.

[34] Anruo Shen, Jingnan Sun, Xiaogang Chen, and Xiaorong Gao. A data-centric and interpretable eeg framework for depression severity grading using shap-based insights. *Journal of NeuroEngineering and Rehabilitation*, 22(1), may 2025. doi: 10.1186/s12984-025-01645-5.

[35] Agnes P. Funk and Mark S. George. Prefrontal eeg asymmetry as a potential biomarker of antidepressant treatment response with transcranial magnetic stimulation (tms): A case series. *Clinical EEG and Neuroscience*, 2008. doi: 10.1177/155005940803900306. URL https://doi.org/10.1177/155005940803900306.

[36] N. Bailey, K. Hoy, C. Sullivan, B. Allman, N. Rogasch, Z. Daskalakis, and P. Fitzgerald. Concurrent transcranial magnetic stimulation and electroencephalography measures are associated with antidepressant response from rtms treatment for depression. *medRxiv*, 2023. doi: 10.1101/2023.02.10.23285794. URL https://www.semanticscholar.org/paper/8278c0b8d2b48ae5ed473c09d6c3f9ba34f53391.

[37] Nicholas Murphy, Amanda J. F. Tamman, Marijn Lijffijt, Dania Amarneh, Sidra Iqbal, Alan C. Swann, Lynnette A. Averill, Brittany OBrien, and Sanjay J. Mathew. Neural complexity eeg biomarkers of rapid and post-rapid ketamine effects in late-life treatment-resistant depression: a randomized control trial. *Neuropsychopharmacology*, 2023. doi: 10.1038/s41386-023-01586-4. URL https://doi.org/10.1038/s41386-023-01586-4.

[38] Ling Guo, Zhuo Zhang, X. Tan, K. Phua, C. Wang, P. Tor, and K. Ang. Resting-state eeg biomarkers of accelerated intermittent theta burst stimulation treatment for depression: a pilot study. In *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2024. doi: 10.1109/EMBC53108.2024.10782112. URL https://www.semanticscholar.org/paper/9d00f9320f2f75435719d9cccc4133e989f8737e.

[39] Yi-Chun Tsai, Cheng-Ta Li, and Chi-Hung Juan. A review of critical brain oscillations in depression and the efficacy of transcranial magnetic stimulation treatment. *Frontiers in Psychiatry*, 2023. doi: 10.3389/fpsyt.2023.1073984. URL https://doi.org/10.3389/fpsyt.2023.1073984.

[40] Paulina Clara Dagnino, Claire Braboszcz, Eleni Kroupi, Maike Splittgerber, Hannah Brauer, Astrid Dempfle, Carolin Breitling-Ziegler, Alexander Prehn-Kristensen, Kerstin Krauel, Michael Siniatchkin, et al. Stratification of responses to tdcs intervention in a healthy pediatric population based on resting-state eeg profiles. *Scientific Reports*, 13(1):8438, 2023.

[41] J. Sheen, F. Mazza, D. Momi, J. Miron, Farrokh Mansouri, Thomas Russell, Ryan Zhou, M. Hyde, L. Fox, Helena Voetterl, E. B. Assi, Z. Daskalakis, D. Blumberger, John D. Griffiths, and Jonathan Downar. N100 as a response prediction biomarker for accelerated 1 hz right dlpfc-rtms in major depression. *Journal of Affective Disorders*, 2024. doi: 10.1016/j.jad.2024.07.131. URL https://www.semanticscholar.org/paper/c0dabdc70010a5fb8ea6f2546d37fe9e83a13071.

[42] Ali Asadi Zeidabadi and Saeid Rashidi. Prediction of rtms treatment response in depression using a frequency-based eeg biomarker. In *International Conference on Computer and Knowledge Engineering*, 2025. doi: 10.1109/ICCKE68588.2025.11273345. URL https://www.semanticscholar.org/paper/fa7b0c55f517ebfbaa6af5be40ba57129e5d87f2.

[43] Andreas Reissmann, Maximilian Rupprecht, B. Langguth, Johanna Rischer, and S. Schoisswohl. Theta-cordance as a biomarker of treatment response to intermittent theta burst stimulation in patients with treatment-resistant depression. *Clinical Neurophysiology*, 2026. doi: 10.1016/j.clinph.2026.2111710. URL https://www.semanticscholar.org/paper/c2c05fd78b73d7ebd82bcbd5cb8af3516e1f5751.

[44] Sarah Ulrich, Else Schneider, Gunnar Deuring, Saskia Erni, Maria de Ridder, Jan Sarlon, and Annette Beatrix Bruhl. Alterations in resting-state eeg functional connectivity in patients with major depressive disorder receiving electroconvulsive therapy: A systematic review. *Neuroscience & Biobehavioral Reviews*, 2025. doi: 10.1016/j.neubiorev.2025.106017. URL https://doi.org/10.1016/j.neubiorev.2025.106017.

[45] Jijomon Chettuthara Moncy, Yong Fan, and Cynthia H.Y. Fu. Prediction of treatment outcome to transcranial direct current stimulation in major depression based on deep learning of eeg data. In *2024 IEEE Conference on Artificial Intelligence (CAI)*, pages

1123–1128, 2024. doi: 10.1109/CAI59869.2024.00201.

[46] Wenyi Xiao, Jijomon C Moncy, Rachel D Woodham, Sudhakar Selvaraj, Nahed Lajmi, Harriet Hobday, Gabrielle Sheehan, Ali-Reza Ghazi-Noori, Peter J Lagerberg, Rodrigo Machado-Vieira, et al. Home-based transcranial direct current stimulation (tdcs) in major depressive disorder: enhanced network synchronization with active relative to sham and deep learning-based predictors of remission. *Personalized Medicine in Psychiatry*, 49:100147, 2025.

[47] Shalini Mahato, Nishant Goyal, Daya Ram, and Sanchita Paul. Detection of depression and scaling of severity using six channel EEG data. *J. Med. Syst.*, 44(7):118, May 2020.

[48] Hao Zhang, Qing-Qi Zhou, He Chen, Xiao-Qing Hu, Wei-Guang Li, Yang Bai, Jun-Xia Han, Yao Wang, Zhen-Hu Liang, Dan Chen, Feng-Yu Cong, Jia-Qing Yan, and Xiao-Li Li. The applied principles of eeg analysis methods in neuroscience and clinical neurology. *Military Medical Research*, 10(1), dec 2023. doi: 10.1186/s40779-023-00502-7.

[49] P. Welch. The use of fast fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15 (2):70–73, jun 1967. doi: 10.1109/TAU.1967.1161901.

[50] Wajid Mumtaz, Likun Xia, Mohd Azhar Mohd Yasin, Syed Saad Azhar Ali, and Aamir Saeed Malik. A wavelet-based technique to predict treatment outcome for major depressive disorder. *PLOS ONE*, 12(2):e0171409, feb 2017. doi: 10.1371/journal.pone.0171409.

[51] Natalia Jaworska, Hongye Wang, Dylan M. Smith, Pierre Blier, Verner Knott, and Andrea B. Protzner. Pre-treatment eeg signal variability is associated with treatment success in depression. *NeuroImage: Clinical*, 17:368–377, 2018. doi: 10.1016/j.nicl.2017.10.035.

[52] Bo Hjorth. Eeg analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology*, 29(3): 306–310, sep 1970. doi: 10.1016/0013-4694(70)90143-4.

[53] Oliver Faust, Peng Chuan Alvin Ang, Subha D. Puthankattil, and Paul K. Joseph. Depression diagnosis support system based on eeg signal entropies. *Journal of Mechanics in Medicine and Biology*, 14 (3):1450035, mar 2014. doi: 10.1142/S0219519414500353.

[54] U. Rajendra Acharya, Vidya K. Sudarshan, Hojjat Adeli, Jayasree Santhosh, Joel E. W. Koh, Subha D. Puthankatti, and Amir Adeli. A novel depression diagnosis index using nonlinear features in eeg signals. *European Neurology*, 74(1-2):79–83, 2015. doi: 10.1159/000438457.

[55] Brian Lord and John J. B. Allen. Evaluating eeg complexity metrics as biomarkers for depression. *Psychophysiology*, 60(8), feb 2023. doi: 10.1111/psyp.14274.

[56] Alessandra Monni, Katherine L. Collison, Kaylin E. Hill, Belel Ait Oumeziane, and Dan Foti. The novel frontal alpha asymmetry factor and its association with depression, anxiety, and personality traits. *Psychophysiology*, 59(11):e14109, may 2022. doi: 10.1111/psyp.14109.

[57] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, jun 2002. doi: 10.1613/jair.953.

[58] Alonso, Esther, Arnott, Stephen R., Atluri, Sravya, Blumberger, Daniel, Brenner, Colleen A., Daskalakis, Zafiris J., Dhami, Prabhjot, Dharsee, Moyez, Evans, Kenneth R., Farzan, Faranak, Frehlich, Matthew, Frey, Benicio N., Kennedy, Sidney H., Kleffner, Killian, Lam, Raymond W., Liotti, Mario, Mcandrews, Mary Pat, Milev, Roumen, Price, Rae, Ravindran, Arun, Rotzinger, Susan, Vila-Rodriguez, Fidel, Wong, and Willy. Standardization of electroencephalography for multi-site, multi-platform and multi-investigator studies: Insights from the canadian biomarker integration network in depression. *Scientific Reports*, 2017. doi: 10.1038/s41598-017-07613-x. URL https://www.research.unipd.it/bitstream/11577/3260320/1/Liotti_SciRep2017.pdf.

[59] Jijomon Chettuthara Moncy, Wenyi Xiao, Rachel D Woodham, Ali-Reza Ghazi-Noori, Hakimeh Rezaei, Elvira Bramon, Philipp Ritter, Michael Bauer, Allan H Young, Yong Fan, and Cynthia H.Y. Fu. Deep learning based treatment remission prediction to transcranial direct current stimulation in bipolar depression using eeg power spectral density. *medRxiv*, 2025. doi: 10.1101/2024.07.16.24310445.

[60] Wenyi Xiao, Jijomon C Moncy, Ali-Reza Ghazi-Noori, Rachel D Woodham, Hakimeh Rezaei, Elvira Bramon, Philipp Ritter, Michael Bauer, Allan H Young, and Cynthia HY Fu. Enhanced network synchronization connectivity following transcranial direct current stimulation (tdcs) in bipolar depression: effects on eeg oscillations and deep learning-based predictors of clinical remission. *Journal of Affective Disorders*, 369:576–587, 2025.

[61] R. Romero Marin, S. Lopez Rodriguez, S. Lakis Granell, E. Buloz Osorio, M. Cabello Toscano, Mikel Urretavizcaya Sarachaga, Maria del Pino Alonso Ortega, Mohit Chopra, J. Solana Sanchez, J. Camprodon, A. Pascual Leone, D. Bartres Faz, Davide Cappon, and G. Cattaneo. Remotely supervised home-based tdcs in treatment-resistant depression: Feasibility, effectiveness and eeg biomarkers of response. *Research Square*, 2026. doi: 10.21203/rs.3.rs-8709698/v1. URL https://www.semanticscholar.org/paper/7ba628353818d20f18ac5ecdccca2bd39bca3d77.

[62] Ian H. Kratter, Christopher W. Austelle, Jennifer I. Lissemore, Masataka Wada, Andrew Geoly, Anna Chaiken, Irakli Kaloiani, Noriah Johnson, Stephanie Wan, Lena Kozyr, Ethan Makarewycz, Brendan L. Wong, Malvika Sridhar, Flint M. Espil, Nick Bassano, Bora Kim, Jarrod Ehrie, Adi Maron Katz, Claudia Tischler, Romina Nejad, Jean-Marie Batail, Angela Phillips, Eleanor Cole, Tiffany J. Ford, Brandon S. Bentzley, Booil Jo, Alan F. Schatzberg, David Spiegel, Cammie Rolle, Gregory L. Sahlem, and Nolan Williams. Stanford neuromodulation therapy for treatment-resistant depression: a randomized controlled trial confirming efficacy, and an <scp>eeg</scp> study providing insight into mechanism of action and a potentially predictive biomarker of efficacy. *World Psychiatry*, 2026. doi: 10.1002/wps.70032. URL https://doi.org/10.1002/wps.70032.

[63] Darcy A. Waller, Linda L. Carpenter, and Stephanie R. Jones. Transient frontal spectral events from eeg predict antidepressant response to sertraline in depression. *Journal of Psychiatric Research*, 2026. doi: 10.1016/j.jpsychires.2026.03.010. URL https://www.semanticscholar.org/paper/6a3e902c05c07569afa93b0bb3e0b484bf519824.

[64] M. Pettorruso, Giorgio Di Lorenzo, Beatrice Benatti, G. dAndrea, C. Cavallotto, Rosalba Carullo, G. Mancusi, O. Di Marco, G. Mammarella, A. Dattilio, Elisabetta Barlocci, Ilenia Rosa, Alessio Cocco, Lorenzo Pio Padula, Giovanna Bubbico, M. G. Perrucci, Roberto Guidotti, A. DAndrea, L. Marzetti, F. Zoratto, B. Dellosso, and Giovanni Martinotti. Overcoming treatment-resistant depression with machine-learning based tools: a study protocol combining eeg and clinical data to personalize glutamatergic and brain stimulation interventions (selectool project). *Frontiers in Psychiatry*, 2024. doi: 10.3389/fpsyt.2024.1436006. URL https://www.semanticscholar.org/paper/169e53dbe54745e1df0495183343bda9d6a8a29e.